

MERN STACK INTERNSHIP – DAY 4 ASSIGNMENT REPORT

Company

Insta Dot Analytics

Submitted By

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FOR SOURCE CODE:

GIT HUB :- <https://github.com/chayan-soni/MERN-Internship>

Objective

The objective of Day 4 was to learn the fundamentals of React.js, understand component-based frontend development, work with React Hooks such as `useState` and `useEffect`, and establish communication between a React frontend and an Express backend using CORS and the Fetch API.

Topics Covered

1. Introduction to React.js

React is a JavaScript library used for building user interfaces. It allows developers to create reusable components and build dynamic web applications efficiently.

Key Concepts Learned

- Component-Based Architecture
- JSX (JavaScript XML)
- Virtual DOM
- Functional Components
- Reusability of UI Elements

2. React Hooks

useState Hook

The useState hook is used to manage and update data within a React component.

Example use cases:

- Storing form input values
- Managing user interactions
- Updating content dynamically

useEffect Hook

The useEffect hook is used to perform side effects such as:

- Fetching data from APIs
- Running code when a component loads
- Synchronizing data with external systems

3. Cross-Origin Resource Sharing (CORS)

Since the React frontend runs on: <http://localhost:5173>

and the Express backend runs on: <http://localhost:5000>

they are considered different origins by the browser.

To allow communication between them, the CORS middleware was installed and configured in the Express server.

Example:

```
const cors = require("cors");  
app.use(cors());
```

Project Overview

A Student Management System was created using React and Express.

The project consists of:

Frontend

- React.js
- Vite
- useState
- useEffect

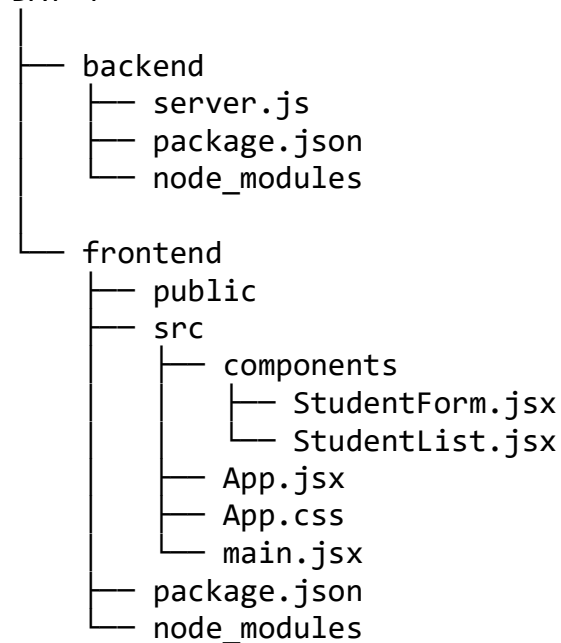
Backend

- Node.js
- Express.js
- CORS

The frontend fetches student data from the backend and displays it dynamically.

Project Structure

DAY 4



Backend Implementation

Installed Packages

```
npm install express cors
```

Server Features

- Express server created
- CORS middleware enabled
- Student API endpoint developed
- JSON data served to frontend

API Endpoint

GET /students

Sample Response

```
[
  {
    "id": 1,
    "name": "Chayan",
    "course": "Computer Science"
  },
  {
    "id": 2,
    "name": "Rahul",
    "course": "Mechanical Engineering"
  }
]
```

Frontend Implementation

React Components Created

StudentForm Component

Purpose:

- Accept user input
- Demonstrate useState functionality
- Display entered text dynamically

StudentList Component

Purpose:

- Fetch student data from backend
- Store fetched data in state
- Render student information dynamically

State Management Using useState

The application uses the useState hook to manage form input values.

Functionality

- User enters text in input field
- State updates instantly
- Updated value appears on screen

Example Output:

You typed: Chayan

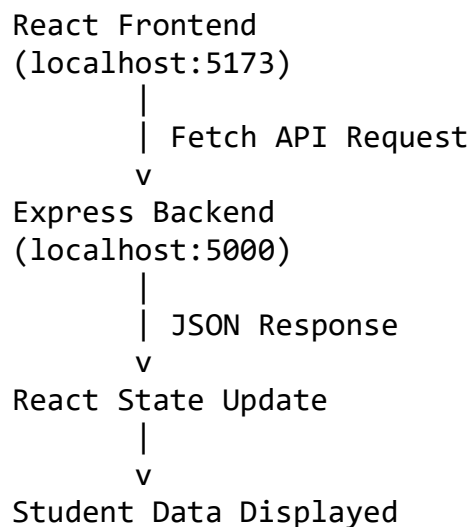
Data Fetching Using useEffect

The useEffect hook was used to fetch data from the Express backend when the component loads.

Process

1. Component loads.
2. useEffect executes.
3. Fetch request is sent to backend.
4. Backend returns JSON data.
5. Data is stored in state.
6. Student list is displayed on screen.

Frontend-Backend Communication Flow



Features Implemented

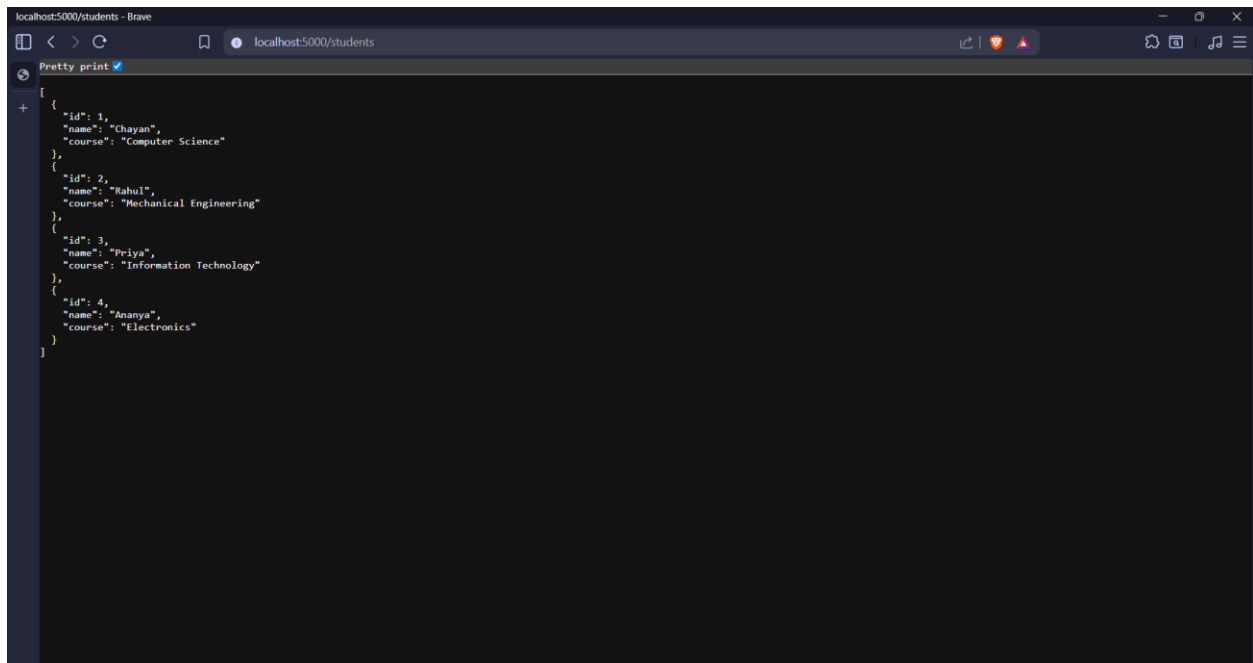
- React project setup using Vite
- Reusable React components
- State management using useState
- API integration using Fetch API
- Data fetching using useEffect
- Express backend setup
- CORS configuration
- Dynamic rendering of student records
- Modular project structure

Screenshots

Screenshot 1

Backend API Response

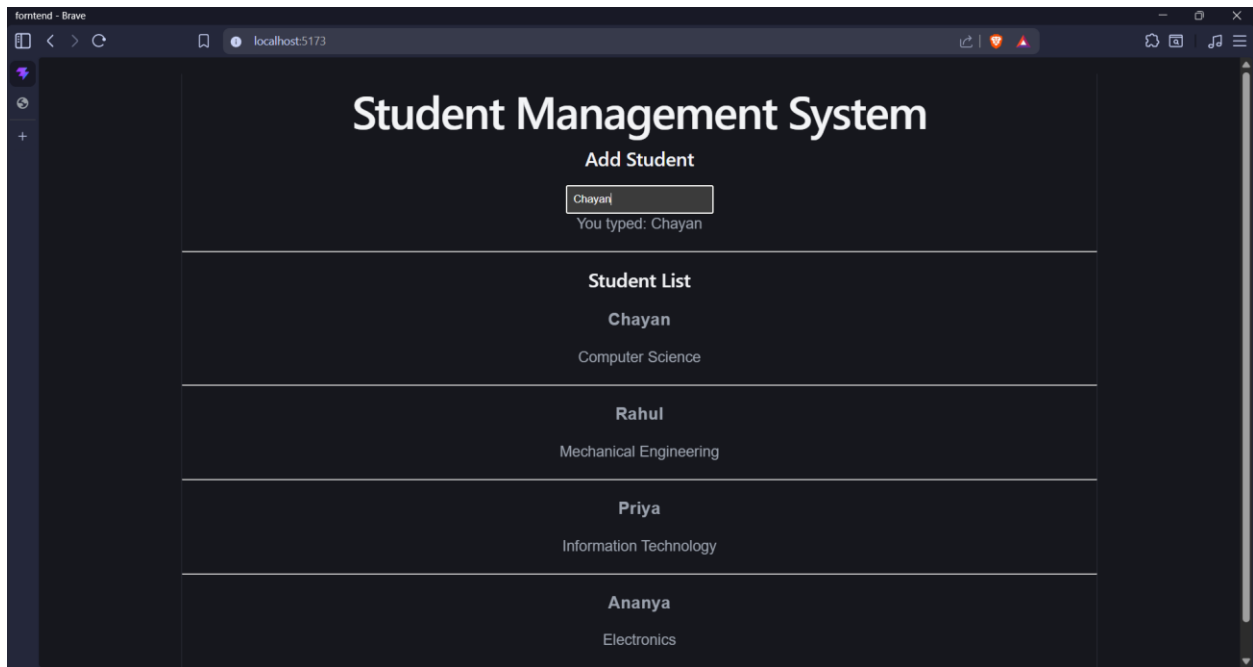
<http://localhost:5000/students>



```
{
  "id": 1,
  "name": "Chayan",
  "course": "Computer Science"
},
{
  "id": 2,
  "name": "Rahul",
  "course": "Mechanical Engineering"
},
{
  "id": 3,
  "name": "Priya",
  "course": "Information Technology"
},
{
  "id": 4,
  "name": "Ananya",
  "course": "Electronics"
}
}
```

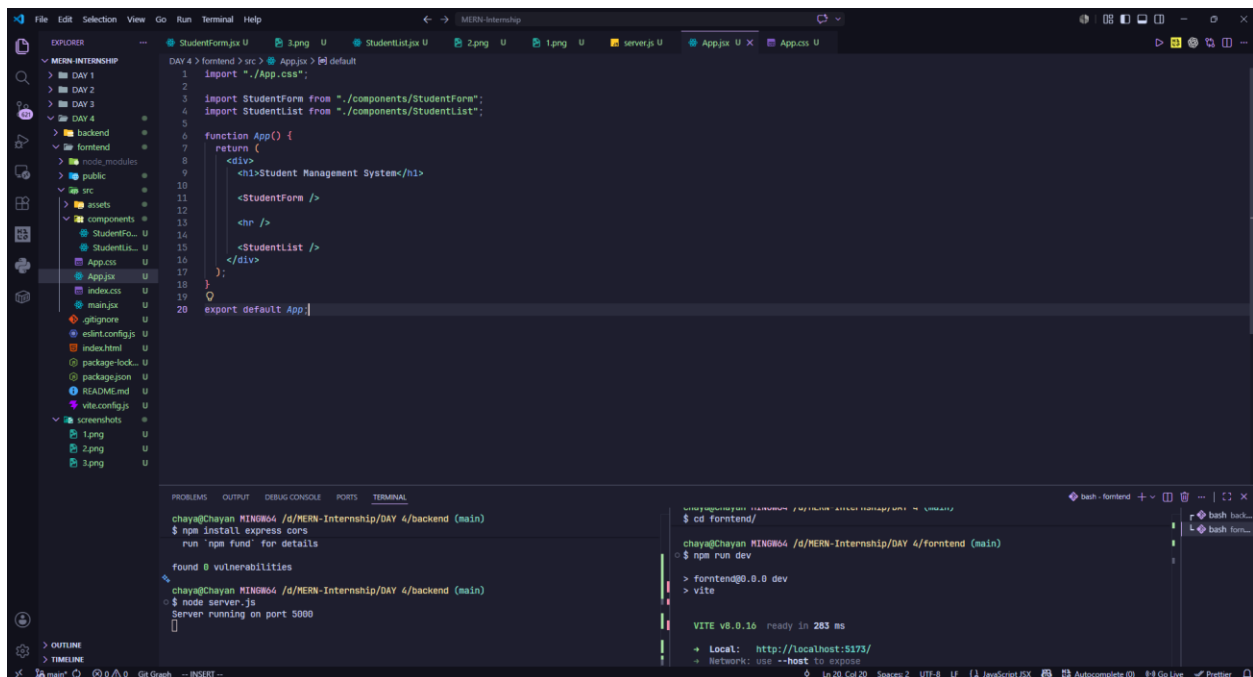
Screenshot 2

React Application Output at localhost:5173



Screenshot 3

Terminal of both frontend and backend successfully running and connected through CORS



GitHub Repository

Repository Link:

<https://github.com/chayan-soni/MERN-Internship>